

CS 403: Algorithm Design and Analysis

Course Code	CS 403
Course Name	Algorithm Design and Analysis
L-T-P-C	3-0-2-4
Prerequisites	CS-3xx (ADSA) or equivalent, or the instructors consent
Intended for	UG
Distribution	Discipline elective for CS and EE
Semester	5th or 6th
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Review of Data Structures.** [3 Lectures]
- **Program Performance:** Time and space complexity, average and worst case analysis, asymptotic notation, recurrence equations and their solution. [3 Lectures]
- **Algorithmic Techniques:** Search techniques (backtracking and bounding), Sorting algorithms - lower bound, sorting in linear time, Greedy algorithms (Huffman coding, knapsack), Divide and conquer - Master theorem, Dynamic programming (0/1 knapsack, Traveling salesman problem, matrix multiplication, all-pairs shortest paths), Randomization, Randomized data structures: Skip Lists, Universal and perfect Hash functions, Backtracking, Branch and bound. [15 Lectures]
 - For each algorithm technique the following is expected: Description of the technique, explanation when an algorithm design situation requires it, examples of algorithms based on this technique, analysis of performance these algorithms.
- **Graph Algorithms:** DFS and BFS, biconnectivity, spanning trees; Minimum cost spanning trees: Kruskals, Prims, and Sollins algorithms; Path finding and shortest path algorithms; Topological sorting; Matching, Network Flows; Bipartite graphs. [6 Lectures]
- **Computational complexity:** Problem classes: P, NP, NP-complete, NP-hard. Reduction. Cooks theorem. Examples of NP-complete problems. [6 Lectures]
- Competitive analysis [3 Lectures]
- **Amortized analysis:** aggregate analysis, accounting, potential method. [3 Lectures]
- **Other topics:** Number theoretic algorithms (GCD, modulo arithmetic, Chinese remainder theorem), string matching algorithms (Rabin Karp algorithm, string matching with Finite State Automata, KMP (Knuth-Morris-Pratt) algorithm, Boyer-Moore algorithm), Strassen's matrix multiplication, FFT, integer and polynomial arithmetic. [3 Lectures]
- **Advanced topics:** Lower-bound techniques: adversary arguments, information-theoretic bounds.

References

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, 3rd Edition, MIT Press, 2009.
2. J. Kleinberg and E. Tardos, **Algorithm Design**, Pearson, 2006.
3. S. Dasgupta, C. H. Papadimitriou, U. V. Vazirani, **Algorithms**, McGraw-Hill, 2006.
4. S. S. Skiena, **The Algorithm Design Manual**, 2nd Edition, Springer, 2008.